

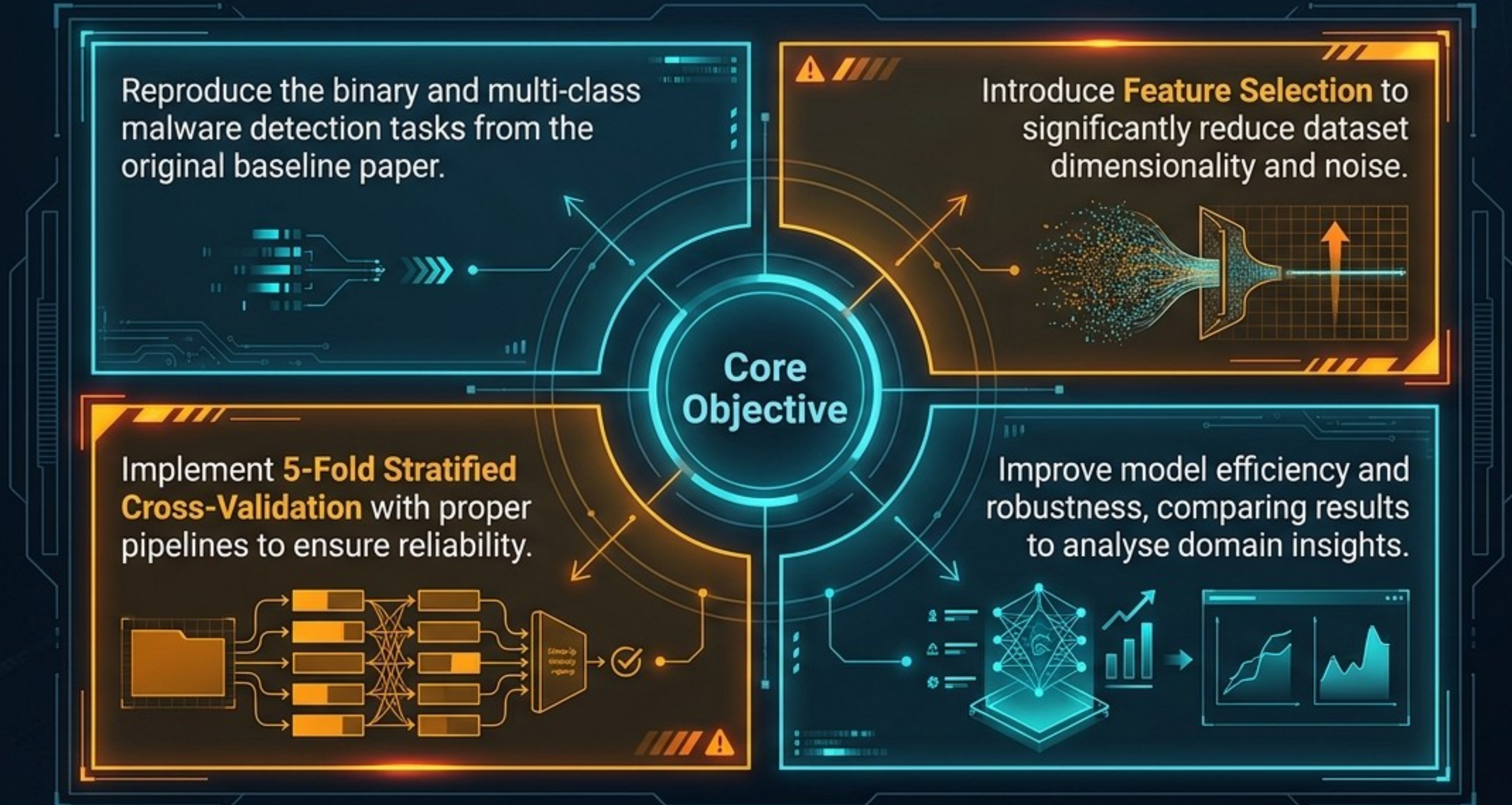
Malware Detection Through Memory Analysis

An Extension with Feature Selection and
Robust Evaluation on the CIC
MalMemAnalysis-2022 Dataset

Girish B Nair

Roll No: 25PG00034

Objectives of the Project



Original Paper Summary

The Baseline Achievements

Dataset: **CIC MaIMemAnalysis-2022**
(58,596 samples)

Features: **52** memory dump features
post-cleaning

Best Model: **XGBoost**

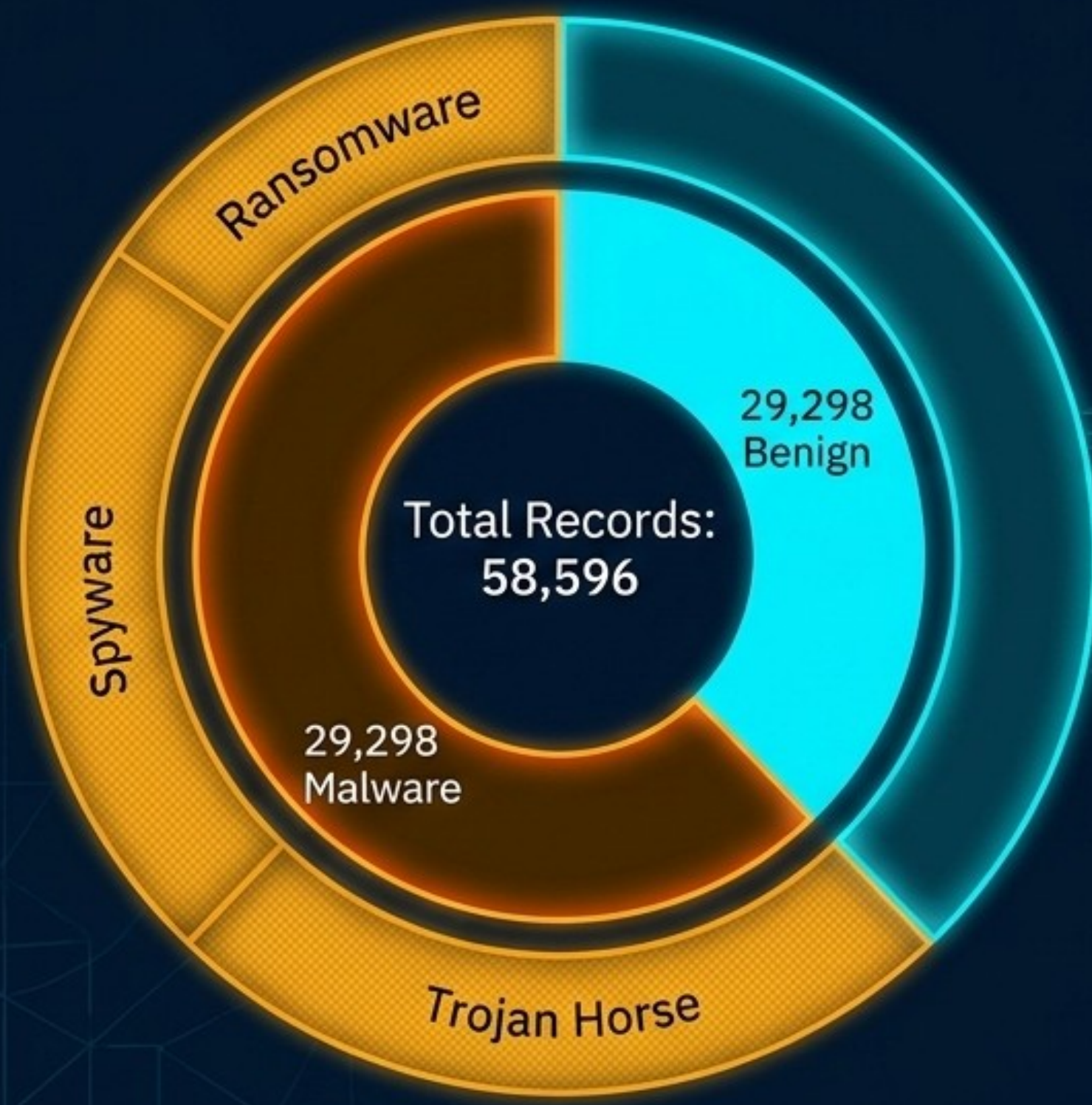
Results:

- Binary: **99.98%** Accuracy & F1
- Multi-class: **87.54%** Accuracy,
81.26% Macro F1

⚠ Methodological Vulnerabilities

- **No feature selection** implemented
- **Limited validation** methodology
(single split)
- High potential for **data leakage**
during scaling

Dataset & Preprocessing



Removed **constant-value** columns



Custom **multi-class** label mapping applied



Stratified **80/20** Train-Test Split enforced



Min-Max Scaling (strictly fitted only on training data)

Key Extension 1 – Feature Selection

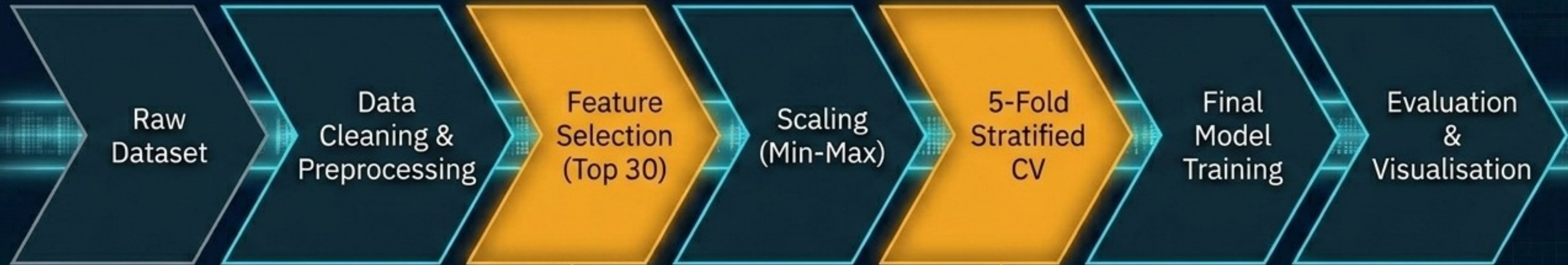


Key Extension 2 – Robust Evaluation



- Implemented 5-Fold Stratified Cross-Validation
- Used Scikit-learn Pipeline architecture
- Ensured Min-Max scaling happens strictly inside each CV fold
- Model Deployed: Random Forest (with hyperparameter tuning)

Proposed Methodology Overview



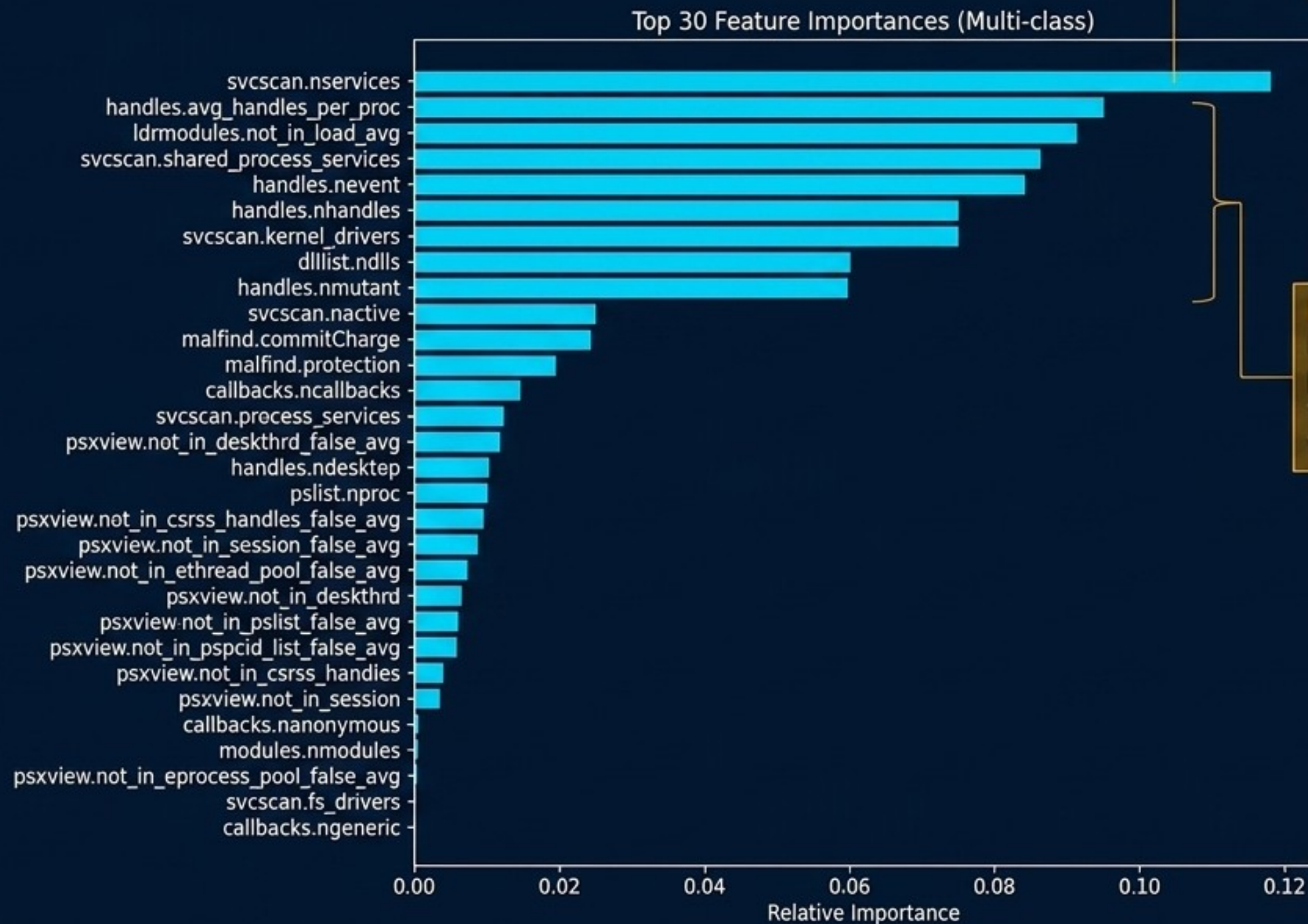
Novel project extensions implemented
to improve baseline reliability.

Results & Comparison

Task	Model	CV Accuracy	Test Accuracy
Binary	Random Forest	99.99	99.99
Multi-class	Random Forest	84.61	76.90
Original	XGBoost	87.22	80.74



Feature Importance & Insights



Primary Indicator:

svcscan.nservices dominates the model's decision-making process.

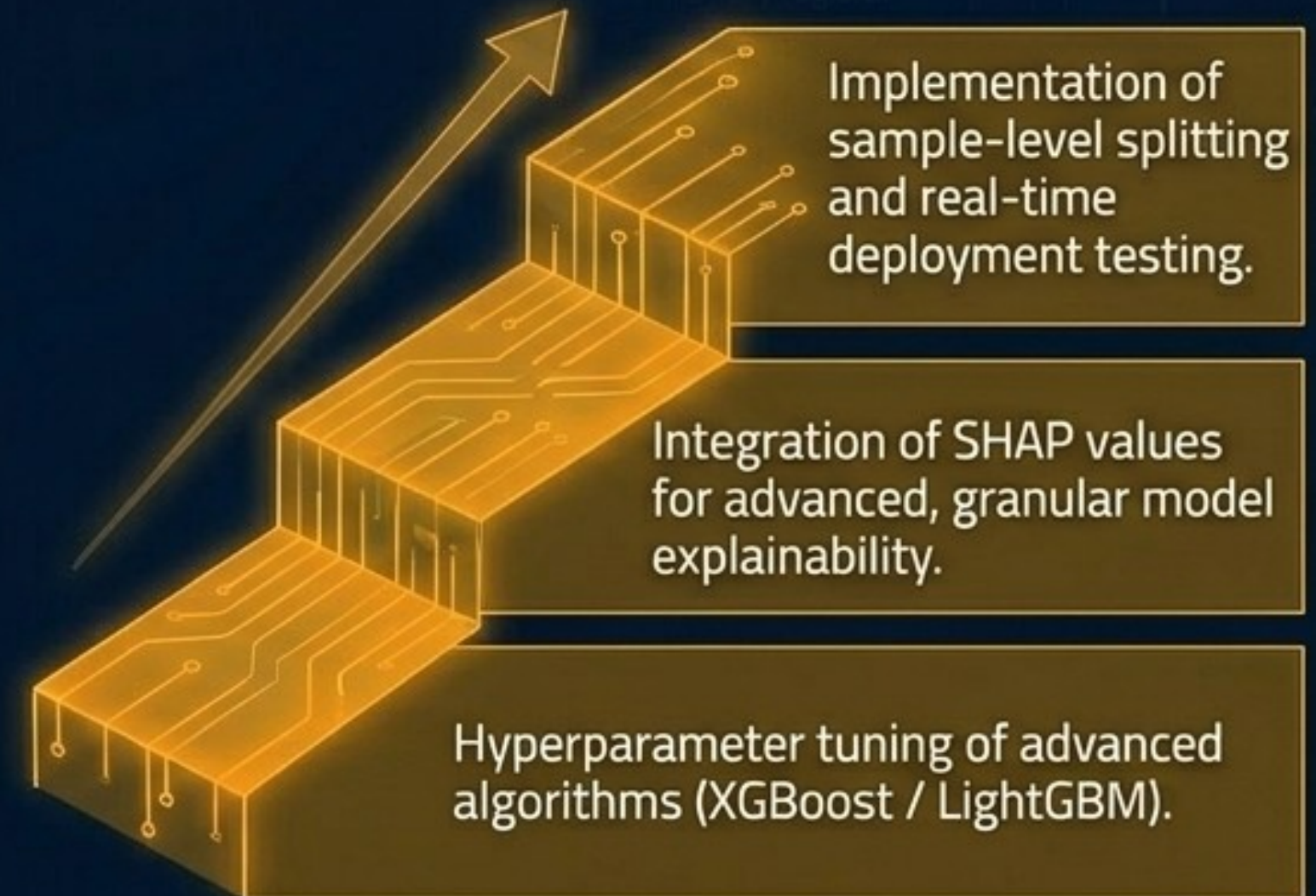
Domain Insight: Memory features related to services, handles, and processes are highly discriminative for malware detection, proving the value of targeted forensic analysis.

Conclusion & Future Work

Conclusions

- ✓ Successfully extended the baseline paper with Feature Selection and Robust Stratified CV.
- ✓ Achieved competitive accuracy while vastly improving model efficiency and reliability.
- ✓ Demonstrated the critical importance of interpretability in cybersecurity ML frameworks.

Future Work



Thank

You